



INTERNATIONAL ORDER OF IT EXPERTS

INTERNATIONAL FOUNDING MOVEMENT

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**WHITE PAPER**

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TOWARDS AN ETHICAL, RESPONSIBLE  
AND SUSTAINABLE DIGITAL WORLD

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FOUNDER & AUTHOR

# White Paper

## *Towards Progressive Recognition of the IT Profession*

INTERNATIONAL ORDER OF IT EXPERTS (IOITE) — INTERNATIONAL FOUNDING  
MOVEMENT — VERSION 3.1 — AUGUST 2026

**Preliminary note on the organization's status.** The International Order of IT Experts (IOITE) is, at this stage, a founding movement backed by an association. It is not a professional order in the legal sense of the term: a professional order is created by law, within the framework of a regulated profession with reserved acts. No private organization can self-attribute this status. The word "Order" is used here as a name in use and as a horizon, not as an acquired legal title.

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## Author's Personal Preface

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I did not write this White Paper because IT lacks recognition. IT is everywhere. It sits at the heart of banks, hospitals, transport, industry, public administrations, and now the artificial intelligence systems that shape a growing share of our collective decisions.

I wrote this White Paper because a paradox seems to me increasingly obvious. We live in a world where software is subject to ever stricter standards, audits, security and compliance requirements. Yet the women and men who design these systems still practice a profession whose boundaries remain largely undefined. Anyone can call themselves an expert. Skill levels vary enormously. Responsibilities are sometimes diluted. Ethical obligations are rarely formalized.

Over the course of my career as an engineer, developer, software architect and then technical lead, I have had the opportunity to work on financial systems, critical platforms and international projects where a single software error could have serious consequences for thousands of people. I discovered a simple truth there: when software becomes infrastructure, the nature of the profession of those who build it changes.

The purpose of this document is not to create an artificial barrier to entering the profession. Nor is it to mechanically replicate the models of other regulated trades. The ambition is more fundamental. It is about opening an international discussion on how our

profession can fully take ownership of its role in modern society — a profession capable of defining its own standards, promoting technical excellence, passing on its knowledge, and collectively assuming its responsibilities.

I am convinced that the coming decades will see IT become one of the most structuring disciplines of our civilization. Digital systems are no longer mere tools; they have become infrastructures of trust. This evolution obliges us.

This White Paper does not provide all the answers. It is above all an invitation to reflection, debate and action. I hope it will contribute, even modestly, to the emergence of a stronger profession, more responsible and more aware of its role in the world.

Yann Deungoué Software Architect Founder of the International Order of IT Experts (IOITE) initiative

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## About the Author

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Yann Deungoué is an engineer and software architect. Over the course of his career — as a developer, architect, and later technical lead — he has designed, built and evolved financial systems, critical platforms and international projects, in environments where the reliability, security and compliance of software are not options but first-order requirements. It is from this hands-on practice, in contact with systems whose failure can affect thousands of people, that the conviction behind this White Paper was born: when software becomes infrastructure, the profession of those who build it changes in nature and calls for responsibility commensurate with its effects.

Based in Switzerland and working in the financial sector, he founded the International Order of IT Experts (IOITE) initiative to open, at an international scale, a discussion on the progressive recognition and structuring of the IT profession. He advocates a measured, non-alarmist approach, focused on critical systems and mindful of preserving the openness, creativity and inclusion of talent from all backgrounds that have made the discipline what it is.

A more complete biographical note, together with a portrait of the author, will be included before the final version is printed.

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## Acknowledgements

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This White Paper owes much to those who, in one way or another, accompanied its maturation. I thank the first readers and reviewers who agreed to examine these pages without complacency, and whose objections mattered as much as their encouragement; the colleagues, engineers, architects, security officers and teachers with whom, over the years, the convictions expressed here were forged; the IT professional community as a whole, whose openness and intellectual generosity remain, in my view, a model to preserve rather than constrain. My gratitude finally goes to my family, whose patience and support made possible work carried out alongside a demanding job. As this document is a first contribution meant to be enriched, the most important acknowledgements are no doubt those I will owe, in future versions, to its contributors to come.

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# Institutional Preface

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Electricity transformed the twentieth century; software is transforming the twenty-first. This sentence, which opens our manifesto, is not a figure of speech. It describes a material shift. In less than two generations, computer code has gone from being a specialized calculation tool to the connective tissue of modern societies. It regulates air traffic, adjusts radiotherapy doses, executes stock market orders, balances power grids, authenticates identities, distributes water and energy, and mediates a growing share of human relations. Where we once saw machines, there are now programs; and behind these programs, professionals whose technical decisions engage — often without anyone measuring it — the safety and interests of a great many people.

This White Paper starts from a simple, verifiable observation: the technical artifacts of IT are among the most rigorously standardized in the world, but the profession that designs them is almost not standardized at all. Network protocols, exchange formats, cryptographic algorithms, and software safety standards are the subject of public, revised, tested and enforceable specifications. Access to the profession that implements them, by contrast, remains in most countries entirely open: no requirement for certified competence, no common enforceable code of ethics, no obligation for continuing education. This imbalance is not necessarily a scandal — it is the product of a fast-moving, creative history. But it has become an anomaly that needs to be named, documented and progressively corrected.

We do not claim to offer a turnkey solution, nor to impose a uniform framework on trades whose diversity is their richness. Building a showcase website does not require the same guarantees as designing the embedded software of a cardiac pacemaker. Our purpose is targeted, graduated, and deliberately modest in its immediate ambitions: to open a collective effort, to establish a shared vocabulary, to propose a competency framework and a code of ethics, and to build the credibility that will, one day, in countries where the legal framework allows it, make a more formal recognition possible. This document is not an endpoint. It is a beginning, offered to the scrutiny of those — academics, security officers, engineers, digital-law specialists, field practitioners — whose feedback will make it better.

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# About IOITE

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The International Order of IT Experts (IOITE) is a founding movement, backed by an association, and not a professional order in the legal sense of the term (see the preliminary note). Its purpose can be summarized in a vision, a mission and a set of values.

## *Vision*

A world in which the professionals who design, develop and operate the digital systems on which our societies depend — health, aviation, energy, finance, administration — practice their profession with a level of competence, ethics and responsibility recognized as commensurate with what is at stake.

## *Mission*

To defend the public interest by promoting responsible, ethical, secure and sustainable IT, and to progressively secure recognition of the IT expert profession as a high-responsibility profession, equipped with:

- a common code of ethics;
- a shared competency framework;
- a continuing-education requirement;
- credible representation before institutions, universities and companies.

## *Values*

Our motto — **"Competence. Ethics. Responsibility."** — sums up the spirit of the movement. We express it through five values that guide our work and our governance.

- **Competence.** We place real, verifiable and continuously maintained mastery above titles, labels and appearances.
- **Ethics.** We treat professional ethics — data protection, honesty, refusal of fraudulent or misleading practices — as a first-order professional requirement, not an optional extra.
- **Responsibility.** We hold that whoever designs a critical system answers for their technical choices, and never hides behind a tool, a machine or an automated system.

- **Independence.** We build our legitimacy, our frameworks and our evaluations independently of commercial interests, in particular software vendors, from whom we intend to remain distinct.
- **Openness.** We advocate a profession that welcomes talent from all backgrounds, transparent in its processes, and we impose requirements only where collective safety justifies them.

### *Name, Structure and International Ambition*

The International Order of IT Experts (IOITE) is the name of the movement: the one that carries the vision set out in this White Paper and brings together the founding members. It should not be confused with the name of the legal entity that hosts it. To avoid any ambiguity with legally constituted professional orders — a question that several reviewers rightly raised with us — the supporting association will be registered under a distinct institutional name, of the form **IOITE Foundation** or **International Institute for IT Responsibility**, with the explicit mention "carrier of the International Order of IT Experts movement." This two-tier structure allows the legal entity to remain neutral and unambiguous, while the movement retains the name that carries its vision and symbolic memory. The definitive legal name will be settled with the support of legal counsel, before any trademark filing; until then, this White Paper systematically uses the full formula International Order of IT Experts — International Founding Movement.

This same requirement for clarity applies to our geographic ambition. IOITE is conceived, from the outset, as an international initiative: this White Paper explicitly calls for "opening an international discussion on how our profession can fully take ownership of its role." But an international ambition does not erase legal reality: the registered office, the legal name of the association and its territorial representations will, as it develops, need to be distinguished — and articulated — country by country, in compliance with local legal frameworks (see also Annex A). The acronym IOITE remains unchanged; it is the institutional framework around it that will be clarified progressively.

# Executive Summary

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This summary condenses into a few pages the essence of the White Paper: the problem, the vision, the proposals, some figures for scale, the roadmap and the expected benefits. It can be read on its own and distributed independently; the body of the document develops and substantiates each point.

## *The Problem*

IT today is one of the most rigorously standardized fields in the technical world. The protocols that run the Internet (the IETF's RFCs), Web standards (W3C), security standards (ISO/IEC 27001, OWASP recommendations) and critical-system safety frameworks (DO-178C for embedded avionics, IEC 62304 for medical devices, ISO 26262 for automotive) tightly govern technical artifacts. Yet in the vast majority of countries, access to the profession that designs these systems remains entirely open: no requirement for certified competence, no common enforceable code of ethics, no obligation for continuing education. Anyone can call themselves an expert.

This paradox creates a blind spot: artifacts are regulated, people are not. Yet the quality and safety of a system depend not only on the compliance of its components; they depend on the judgment, trade-offs and integrity of the professionals who assemble them. This is a collective blind spot, all the more concerning because the effects of software on the world have changed scale.

## *Why Now*

Three concurrent transformations make this gap more problematic than it was twenty years ago:

- **Generative artificial intelligence** is transforming the very nature of software work and shifting value away from writing code toward the ability to specify, verify and take responsibility for a system. A machine can propose; a professional must answer for it.
- **Cybersecurity** has become a matter of collective safety: hospitals, local governments, energy operators and supply chains are now among the targets. The regulatory movement is accelerating (NIS2, DORA, Cyber Resilience Act); the structuring of the profession, meanwhile, lags behind.

- **Global economic and social pressure** (price competition, precarious employment, the proliferation of product-bound commercial certifications) blurs the picture and makes it difficult, for employers as well as public authorities, to tell genuine expertise apart from its imitation.

## *The Vision*

Our thesis can be stated in one sentence: software has become a matter of public interest, comparable in its effects to energy or transport infrastructure, and this reality must be reflected in how the profession that produces it is organized.

We do not propose to regulate all digital trades uniformly — building a personal website does not require the same guarantees as designing embedded medical software. We advocate a targeted, progressive approach, summed up by one guiding principle:

Anyone assuming technical responsibility for a critical system should meet minimum requirements of competence, continuing education and professional ethics.

## *Who Is an IT Expert?*

This question has been put to us directly, and we owe a straightforward answer rather than leaving it unaddressed. An IT expert is not defined by a diploma alone, by seniority, or by mastery of a commercial product. Their expertise results from a verifiable combination of knowledge, experience, demonstrated achievements, responsibility actually exercised, judgment in the face of complex situations, continuing education and ethical commitment. It is this definition, operationalized by the Competency Framework (see Section 10), that distinguishes durable professional expertise from mere apparent productivity or commercial certification — and that addresses both the incidents mentioned above and the "coding at school" paradox: knowing how to write code, even learned young, is not the same as bearing this responsibility.

## *What We Propose to Build*

1. **A competency framework and expertise-level scale**, transparent and public, describing what an IT expert is according to a graduated scale (Practitioner, Engineer, Architect, Expert, Senior Expert, Fellow).

2. **A common code of ethics**, inspired by high-responsibility professions: data protection, refusal of fraudulent practices, disclosure of conflicts of interest, documentation of technical decisions.
3. **A certification framework independent of commercial vendors**, built with academic partners, to distinguish durable professional competence from mere mastery of a product.
4. **A continuing-education requirement**, modeled on the healthcare professions, broken down by major domain (AI, cybersecurity, cloud, regulation, ethics, digital sustainability).
5. **An independent observatory** regularly publishing the real state of the profession.

### *Some Figures for Scale*

Precise data on the IT profession are scattered and methodologically fragile; we therefore stick to orders of magnitude, flagging their estimated nature.

- According to commonly cited professional estimates, the number of software developers worldwide runs into the **tens of millions**, and is growing year on year. In most countries, none of these professionals is bound to a title or an enforceable code of ethics in order to practice.
- The annual global cost of cybercrime is frequently estimated in the **trillions of dollars**, with significant uncertainty depending on the scope and methods used. The upward trend, however, is not in dispute.
- The artificial intelligence technology market is experiencing **rapid growth**, in double digits according to most sector analyses, which correspondingly increases the surface of decisions entrusted to automated systems.
- Major software incidents illustrate telling orders of magnitude: a deployment failure cost a financial firm **around \$440 million in under an hour** in 2012; a faulty security-software update caused, in 2024, an outage affecting **around 8.5 million devices** worldwide within hours, according to the operating-system vendor concerned.
- The **digital dependence** of societies — a handful of cloud providers, a handful of universal libraries, a handful of components deployed identically on millions of machines — concentrates risk: a failure at a single point can now propagate on a planetary scale.

These figures measure the scale of the problem; they do not show that a professional organization would mechanically reduce cybercrime — no organization can stop a determined attacker from acting. What they do show is that the competence, continuing

education and responsibility of professionals have become essential components of digital resilience: it is on this more modest, more honest register — rather than a promise of mechanical reduction — that our expected contribution lies.

### *The Roadmap*

We progress through sequential milestones rather than calendar deadlines: legal and identity lock-in; minimum viable documentary base; public presence; complete corpus (ethics, framework, charters); first circle of credible contributors; broader recognition. Each milestone is a condition for the next. Recognition cannot be built before the corpus, nor the corpus before a shared vocabulary.

### *Expected Benefits*

For **professionals**: recognition of genuine competence, protection against a race to the bottom, legible career benchmarks. For **employers and clients**: reduced information asymmetry, safer hiring, recognition for teams that invest in their competence. For **society**: greater quality, security, resilience and trust in the systems on which health, finance, energy and public services depend.

### *What We Are Not*

We are not a professional order in the legal sense; we do not have the power to create reserved acts or to condition access to the profession, and no private organization can self-attribute this. We are a founding movement, backed by an association, building a corpus. If this corpus acquires sufficient legitimacy, it may serve as a basis for a future discussion with universities, companies and, where appropriate, public authorities, in countries where the legal framework allows it.

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# 1. Why This Document, Why Now

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One might object that the debate on the professionalization of IT is an old one, and that it has never succeeded. That is true, and it is precisely why it must be taken up afresh. Conditions have changed. Three concurrent transformations make today's gap more problematic than it was twenty years ago.

The first is the arrival of generative artificial intelligence in software production itself. Tools capable of writing, completing and correcting code are profoundly altering the nature of the work and the definition of competence. They shift value: less toward writing lines of code, more toward the ability to specify, verify, arbitrate and take responsibility for a system. This shift makes it all the more necessary to have a framework that distinguishes genuine expertise from mere apparent productivity, and that recalls that a machine can propose, but a professional must answer. Far from disappearing, the question of human responsibility becomes central.

The second is the transformation of cybersecurity into a matter of collective safety. Cyberattacks no longer target only commercial data; they strike hospitals, local authorities, energy operators, supply chains. The WannaCry ransomware, in 2017, disrupted part of the British healthcare service; that same year, the NotPetya attack spread well beyond its initial targets, notably paralyzing the systems of shipping company Maersk. These episodes showed that a software failure can have system-wide effects, out of all proportion to the initial incident. Lawmakers have understood this: the European Union has adopted frameworks such as the NIS2 directive for the resilience of essential infrastructure, the DORA regulation for the financial sector, and the Cyber Resilience Act for the security of digital products. The regulatory movement is accelerating; the structuring of the profession, meanwhile, lags behind.

The third transformation is economic and social. The globalization of IT work has created price pressure and job insecurity unknown to comparable regulated professions. The proliferation of commercial certifications, often tied to a vendor and its products, blurs the picture: they attest to mastery of a tool, rarely to a transferable, durable professional competence. The result is an information asymmetry that is hard for employers, clients and public authorities to correct: how, in an opaque market, can one tell genuine expertise from its imitation? A profession that evolves as fast as medicine, but that has neither its code of ethics nor its continuing-education requirement, exposes itself to a double loss: of quality for society, of recognition for its members.



This document exists to set out these observations in an orderly way, to distinguish facts from opinions, and to propose a path forward. It is addressed first to the profession itself, then to academic institutions and companies, and finally — but only once credibility has been patiently built — to public authorities.

## 2. A Brief History of the IT Profession

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Understanding why IT is not structured like a profession requires going back to its youth. Unlike medicine or law, whose professional bodies formed over centuries, IT is younger than a human lifetime. The first programmable general-purpose computers date from the 1940s; the discipline built itself as it went, carried by a succession of breakthroughs rather than slow institutional sedimentation.

Its founding figures come from different worlds: mathematicians, physicists, electronics engineers, logicians. Alan Turing formalized the notion of computability as early as 1936; Grace Hopper developed one of the first compilers and contributed to the birth of human-readable languages, opening the way to a computing no longer reserved for machine designers. As hardware became commoditized, the center of gravity shifted toward software — and with it, complexity. By the late 1960s, practitioners noted that producing reliable programs was far harder than expected: projects overran their budgets, accumulated defects, sometimes failed outright. It was from this diagnosis that, at a NATO conference held in Garmisch in 1968, the phrase "software crisis" emerged, along with the call for genuine software engineering: applying to code production the methodical rigor of engineering.

The ambition was sound, but it ran into a tension that has never been resolved. Classical engineering rests on stable physical laws, characterized materials, proven safety margins. Software, by contrast, is immaterial, infinitely malleable, and its complexity grows without the physical constraints that bound a bridge or an engine. This plasticity is its strength — it explains the extraordinary speed of digital innovation — but it has also fostered a professional culture reluctant to standardize people. Individual talent, self-teaching, the ability to learn fast have always been valued there more than a diploma or a title. Many of the field's most influential contributors never followed a formal course of study in computer science. This openness is a precious asset, and we do not intend to give it up.

It must be said clearly, because it shapes our proposal: the absence of a barrier to entry has been an engine of innovation and inclusion. It has allowed talent from varied backgrounds to contribute, and it has avoided the entrenched positions that lock certain closed professions in place. Our thesis is not that this freedom was a mistake. It is that, within a precise scope — that of critical systems — the social cost of a total absence of requirements has become greater than the benefit of unlimited openness. The profession has matured; its effects on the world have changed scale. It is time for its structuring to catch up, carefully and without disavowal, with this transformation.

### 3. Software as Critical Infrastructure

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That software can kill, ruin or paralyze is not a theoretical hypothesis. It is a documented fact, illustrated by a series of incidents that have become textbook cases, studied in engineering curricula around the world. Recalling them is not sensationalism: they constitute empirical proof that certain software systems do indeed belong to the category of critical infrastructure, in the sense of our glossary — systems whose failure can result in serious consequences for the safety of people, the economy or the functioning of essential infrastructure.

In **healthcare**, the Therac-25 radiotherapy machine, in the mid-1980s, delivered massive radiation overdoses to several patients, causing deaths and serious injuries. The investigation showed that the cause was not a faulty part but a set of software defects — a race condition between tasks, excessive trust placed in the software after hardware interlocks were removed, insufficient handling of error states. Forty years on, this case remains the reference for teaching that a software defect can have lethal consequences, and that the safety of a system is never reducible to that of its hardware components.

In **aviation**, the maiden flight of Ariane 5, in 1996, ended in the destruction of the launcher a few seconds after liftoff. The cause: the reuse of a software module inherited from Ariane 4, in which the conversion of a floating-point value to a shorter integer format caused an unhandled overflow. More recently, the two Boeing 737 MAX crashes, in 2018 and 2019, cost 346 lives; investigations implicated an automatic trim-compensation system (MCAS) that relied on a single data source and whose behavior had not been properly documented or communicated to flight crews. These tragedies illustrate a recurring lesson: it is not always the most sophisticated algorithms that fail, but implicit assumptions, specification gaps and poorly defined areas of responsibility.

In **finance**, the firm Knight Capital lost around \$440 million in under an hour, in 2012, following the faulty deployment of automated trading software that placed erroneous orders at an uncontrollable rate — an incident that nearly brought down the company. In **energy**, the major blackout that hit northeastern North America in 2003, cutting power to tens of millions of people, was aggravated by a software bug in a control center's alarm system, which prevented operators from noticing the deteriorating situation in time. Finally, in 2024, a simple faulty update to a widely deployed security software caused, according to the operating-system vendor concerned, an outage affecting around 8.5 million devices worldwide within hours — grounding planes, disrupting hospitals, banks and administrations. This episode made tangible a reality experts have long called systemic dependence. A single component, present everywhere, becomes a planetary point of failure.

What links these incidents, across such different sectors, is that none of them was inevitable. Each implicates chains of human decisions: architectural choices, schedule trade-offs, insufficient testing procedures, diluted responsibilities. This is precisely what the disciplines governing critical systems seek to discipline — and it is here that the gap we will return to opens up, between the rigor imposed on artifacts and the total freedom left to people.

That said, one must guard against too hasty a reading of these episodes, which would amount to equating major failure with individual incompetence. None of the cases cited above demonstrates a lack of technical competence on the part of the professionals involved: the investigations implicate, depending on the case, specification gaps, architectural choices questionable only in hindsight, insufficient testing, economic constraints, failing governance, or responsibilities diluted across several teams — rarely a simple error that better individual competence alone would have sufficed to prevent. It is precisely this observation that grounds our thesis: technical competence, even genuine, is not enough; it must be embedded in a collective framework of governance, ethics and the ability to flag risks — without which the best individual competence remains powerless against organizational failures.

## 4. New Risks: AI, Cybersecurity, Systemic Dependence

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To the classic risks of critical software are now added risks of a new nature, calling for competencies and vigilance that the profession has not yet collectively acquired.

The first stems from the **spread of artificial intelligence** into decision-making systems. Machine-learning models are not programmed in the classical sense: they are trained on data, and their behavior emerges from that training rather than from an explicit specification. This produces novel properties — opacity of decisions, sensitivity to biases present in the data, unpredictable behavior outside the training domain, difficulty guaranteeing a reproducible result. When such systems are involved in screening job applications, granting credit, assisting medical diagnosis or moderating content, the question of responsibility becomes acute. The notion of responsible AI, as we define it — transparency, fairness, safety and human responsibility — is not a nicety: it is an engineering requirement. Lawmakers are beginning to enshrine it in law, as with the European regulation on artificial intelligence, which graduates obligations according to risk level. But a piece of legislation does not train practitioners: professionals capable of implementing it with discernment are still needed.

The second risk is **cybersecurity**, now inseparable from system design itself. Software vulnerabilities are no longer isolated incidents but a structural phenomenon. The Heartbleed flaw, discovered in 2014 in a ubiquitous cryptographic library, exposed the data of millions of servers; the Log4Shell vulnerability, in 2021, affected a logging library so widely used that it took months to catalogue all its uses. These episodes reveal a deep dependence on shared software components, often developed and maintained by a handful of volunteers, on which an immense share of the digital economy nonetheless depends. The attempted introduction of a backdoor into a widely distributed compression utility, narrowly foiled in 2024, showed that this software supply chain constitutes an attack surface in its own right. Securing a system is no longer just about protecting its perimeter: one must understand, trace and control the whole of its dependencies.

The third, cross-cutting risk is **systemic dependence** itself. The pooling of infrastructure — a handful of cloud providers, a handful of universal libraries, a handful of security software deployed identically on millions of machines — brings enormous efficiency gains, but concentrates risk. A failure at a single point now propagates on a global scale, as shown by the 2024 global outage mentioned above. This issue connects with that of digital sovereignty: the ability of a state, an organization or an individual to control its infrastructure and technology choices without excessive dependence on third parties. We

do not approach this issue ideologically, but from the standpoint of resilience: a system whose components and points of concentration are not under one's control is a system whose risk is not under one's control.

These three risks share a common denominator: they exceed the competence of any single individual and demand a shared professional culture — methods, ethics, continuous updating of knowledge. This is exactly what an organized profession can provide.

## 5. Green IT and Digital Sustainability

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An issue long neglected is now becoming pressing: the environmental footprint of digital technology. IT was initially seen as a "clean," immaterial technology, in contrast to heavy industry. This image is misleading. Behind software's apparent immateriality lie energy-hungry data centers, global telecommunications networks, and above all a substantial flow of equipment whose manufacture consumes scarce resources and significant energy.

Available estimates place the share of digital technology at around a few percent of global greenhouse-gas emissions — an order of magnitude often compared to that of air travel — and this share is growing. We remain cautious about precise figures, which vary according to methodology and scope, but the underlying trend is not in dispute: demand for computing is growing rapidly, driven in particular by the rise of artificial intelligence, whose training and operation of large models are especially energy- and hardware-intensive. One point commands consensus among serious analyses: the predominant share of the impact comes from the manufacture of devices and equipment, more than from their electricity consumption in operation alone. This means that extending equipment lifespans and exercising restraint in renewing hardware weigh at least as much as the energy efficiency of data centers.

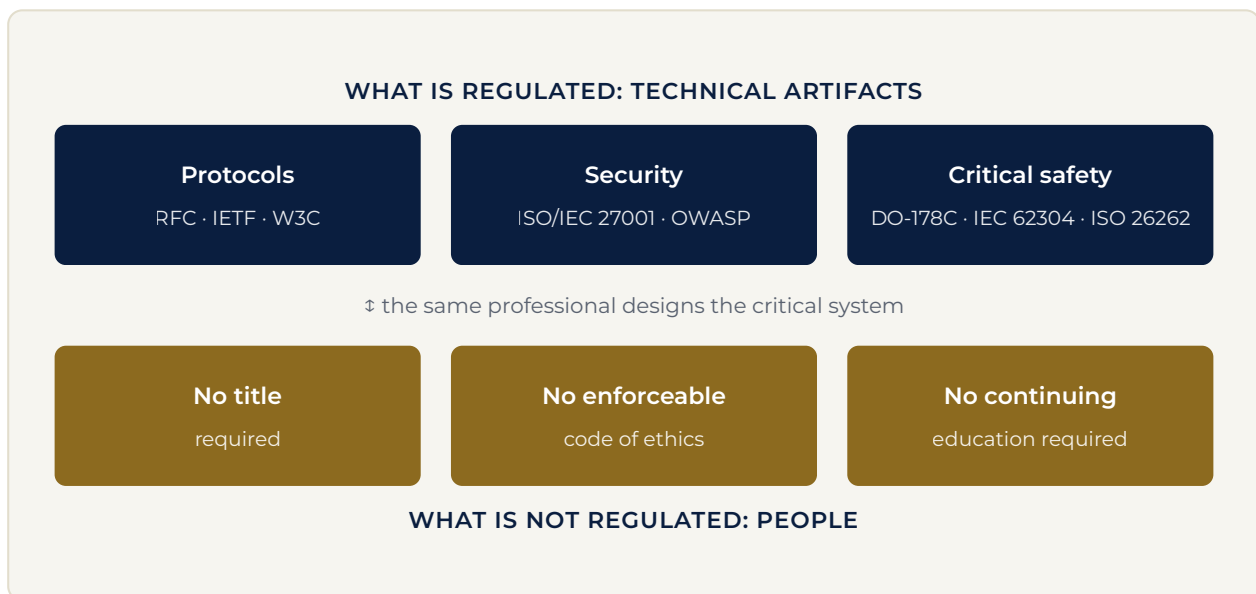
This is where professional responsibility comes into play, because many seemingly technical decisions have direct environmental consequences. A poorly designed software architecture can needlessly multiply computing and storage needs. Software whose requirements grow without necessity — sometimes called "bloatware" — pushes toward the premature replacement of still-functional hardware, generating avoidable electronic waste. Conversely, lean design, optimized processing, right-sized infrastructure and thoughtful choice of computing locations can significantly reduce the footprint, often while also improving performance and cost. Green IT, as we define it — the set of practices aimed at reducing the environmental footprint of digital systems — is therefore not a separate domain reserved for specialists: it is a cross-cutting dimension of the profession, on a par with security.

Frameworks and communities are emerging to support this requirement, such as work on software eco-design, application footprint measurement, and principles championed by collectives dedicated to sustainable software. Public authorities and regulators are also taking up the issue, with growing obligations to measure and reduce digital footprints. We believe digital sustainability must appear explicitly in the competency framework and in the profession's code of ethics: not as an imposed constraint, but as a marker of professional quality. Designing an efficient system also means designing a lean one.

## 6. The Paradox: Standardized Systems, an Unregulated Profession

We come here to the heart of our argument. IT presents a striking singularity: few fields of human activity have standardized their technical artifacts so extensively, while leaving access to the profession that produces them so unstructured.

The diagram below summarizes this imbalance: on one side, a dense, enforceable stack of technical requirements; on the other, an almost total absence of requirements on people — even though it is very often the same professional.



Consider first the scale of technical standardization. Digital communications rest on thousands of public, revised specifications — the RFCs published by the IETF for Internet protocols, W3C standards for the Web, ISO and IEC standards for countless aspects of software and data. Security has its own frameworks: the ISO/IEC 27001 standard for information security management systems, OWASP recommendations for application security, public vulnerability catalogues and their scoring systems. High-criticality domains go further still: the DO-178C standard governs embedded avionics software and imposes rigorous traceability between requirements, code and tests; the IEC 62304 standard governs the software lifecycle of medical devices; the ISO 26262 standard addresses the functional safety of automotive systems; the generic IEC 61508 standard underlies an entire family of requirements for safety-related systems. In these sectors, a software component cannot be put into service without meeting detailed, auditable, enforceable requirements.



Now consider the other side of the ledger. In the vast majority of countries, no qualification is legally required to design, develop or operate most of these systems. The same person can, without any recognized title or enforceable code of ethics, write the code that will process health data, orchestrate financial transactions, or control industrial equipment. Where doctors, lawyers, pharmacists or accountants practice a regulated profession — access to which is conditioned on a title, a diploma or registration, and governed by ethics and discipline — IT remains, in the sense of our glossary, a profession that is neither regulated nor even, everywhere, organized.

The paradox can then be stated simply: we scrupulously regulate artifacts, and not at all the people who produce them. Yet the quality and safety of a system depend not only on the compliance of its components with standards; they depend on the judgment, trade-offs and integrity of the professionals who assemble them. Technical standards presuppose competent practitioners to apply them with discernment — but nothing guarantees this competence, nor maintains it over time. This is a collective blind spot.

Does this mean IT is the only profession in this situation? No, and it would be dishonest to suggest so. Management consulting, private security outside regulated acts, data science, part of compliance consulting, or certain construction trades outside reserved acts also combine dense technical standardization with widely open professional access. IT's singularity therefore does not come from its isolation, but from the combination, to this degree, of four factors: its presence across all sectors of activity, the speed of its evolution, its capacity to propagate globally, and consequences that can be physical, economic and social all at once. Few professions combine these four traits simultaneously; it is this combination, more than the absence of any comparison, that justifies our approach.

One must guard against an excessive conclusion. This paradox does not mean the entire trade should be regulated, nor that barriers should be erected where openness has proven its worth. It means a link is missing: between the technical standard, which applies to the object, and general law, which applies to everyone, there is no intermediate professional framework specific to critical systems. It is this missing link that we propose to build — starting by documenting it and making it desirable, rather than claiming to impose it.

## 7. Existing International Models

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We are not starting from a blank page. Other professions, and IT itself in some of its components, have built forms of structuring from which we can draw inspiration — carefully distinguishing what belongs to a legal status, which we do not have, from what belongs to an authority earned through quality, which we can aim for.

The first model is that of **classical professional orders**: doctors, pharmacists, architects, chartered accountants. What they have in common is combining a legally protected title, an enforceable code of ethics, a disciplinary body and a continuing-education requirement. These professions demonstrate that it is possible to reconcile a high level of requirements with lasting public trust. We take from them their grammar — ethics, verified competence, continuing education, discipline — without claiming their legal status, which presupposes a law we have no power to enact.

The second, more directly relevant model is that of **regulated engineering**, of which the Canadian example is instructive. In Quebec, the Ordre des ingénieurs du Québec governs the title of engineer, protected by law, with reserved activities in certain fields and a disciplinary mechanism. The Canadian tradition has even forged a ritual — the Iron Ring given to graduating engineers — to symbolically anchor an awareness of responsibility. Software engineering has been partially integrated into this framework, depending on the province. In the United States, a Professional Engineer license in software engineering even existed, before the corresponding exam was discontinued for lack of sufficient candidates — a sign that transposing the physical-engineering model onto software still runs into cultural and practical resistance. These experiences, including their limitations, are for us a valuable source of learning: they show both the way forward and its obstacles.

The third model is probably the most inspiring for our immediate situation: that of **large technical organizations without order status**. IEEE and ACM, leading learned societies in electronics and computing, have jointly developed a software-engineering code of ethics that carries intellectual authority well beyond any legal framework. The IETF and W3C produce the standards that run the Internet and the Web without holding any binding power: their authority rests entirely on the quality of their work, the transparency of their processes and the voluntary buy-in of the community. These organizations prove that strong legitimacy can be built without legal status, through merited recognition alone. This is precisely the trajectory we aim for with IOITE in its first phase: to become a reference through quality, even before becoming — perhaps one day — an institution recognized by law.

From these models, we draw a guiding line: borrow from the orders their ethical rigor and their culture of responsibility; borrow from regulated engineering its gradation of levels and its notion of sensitive activities; borrow from the large technical organizations their method — authority through quality, openness and transparency, rather than through compulsion.

## 8. Our Thesis and Our Proposal

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Our thesis can be stated in one sentence: software has become a matter of public interest, comparable in its effects to energy or transport infrastructure, and this reality must be reflected in how the profession that produces it is organized. When a computer system controls an aircraft, a hospital, a bank or a power grid, an error can have major human and economic consequences. Society is entitled to expect, from those who bear this responsibility, a level of competence, ethics and vigilance commensurate with what is at stake.

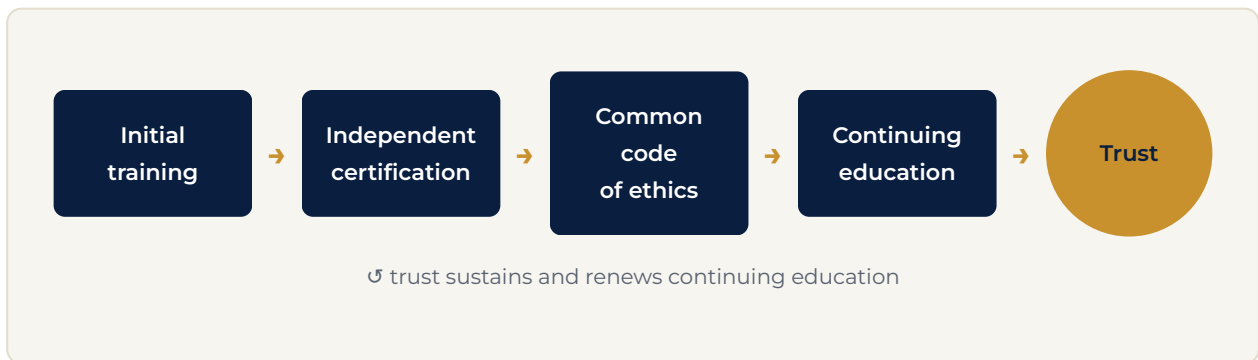
From this thesis follows a proposal whose targeted and progressive nature we want to emphasize from the outset, since this is what distinguishes it from past attempts. We do not propose to regulate all digital trades uniformly. Such an ambition would be both unrealistic and harmful: it would stifle innovation, create entrenched positions, and rightly meet the profession's refusal. Our guiding principle is as follows:

Anyone assuming technical responsibility for a critical system should meet minimum requirements of competence, continuing education and professional ethics.

The key word is critical, in the precise sense of our glossary. Building a personal website, an independent video game, or an internal tool with no consequences for safety or the economy requires none of the guarantees we call for. Designing the embedded software of a medical device, the orchestration of a power grid, or the transactional core of a banking system falls under a different regime of responsibility. It is on this narrower scope that the requirement should bear — a scope which the Competency Framework will have the task of delimiting rigorously, avoiding both excessive extension and complacent restriction.

What we propose to build breaks down into five workstreams, developed in the following sections. First, a competency framework and expertise-level scale, transparent and public, describing what an IT expert is according to a graduated scale. Second, a common code of ethics, inspired by high-responsibility professions. Third, a certification framework independent of commercial vendors, built with academic partners, to distinguish durable professional competence from mere mastery of a product. Fourth, a continuing-education requirement, modeled on the healthcare professions, which recognizes that IT expertise ages quickly and must be maintained. Fifth, an independent observatory regularly publishing the real state of the profession.

These five workstreams are not merely juxtaposed: they form a coherent value chain, in which each link produces the raw material for the next, and the whole converges on a single objective — trust. Initial training feeds certification; certification is articulated with a common code of ethics; ethics is sustained through continuing education; and it is from this whole that trust in critical systems arises, to the benefit of professionals and society alike.



We want to be clear about the purpose of this proposal, since the question has rightly been put to us: structuring a profession can be seen as an attempt to limit competition, protect a job market, or restrict access to the trade — a guild-like reflex that professional orders have sometimes embodied in the past. That is not our logic, and we set out this hierarchy unambiguously:

1. protection of the public;
2. trust in the systems our societies depend on;
3. professional quality;
4. recognition of experts.

The recognition and protection of professionals — clearer career benchmarks, the legitimate refusal of dangerous instructions, protection against the misappropriation of expertise — are a desirable consequence of this structuring, not its primary purpose. It is this hierarchy, and not its reverse, that distinguishes our approach from a guild-protection scheme.

It bears repeating what we are not. We are not a professional order in the legal sense; we do not have the power to create reserved acts or to condition access to the profession. We are a founding movement, backed by an association, building a corpus. If this corpus acquires sufficient legitimacy, it may serve as a basis for a future discussion with universities, companies and, where appropriate, public authorities, in countries where the legal framework allows it. We do not seek to erect an artificial barrier or a monopoly, but to improve quality, transparency and trust — for the benefit of professionals as much as of society.

## 9. Proposed Governance

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An organization that claims to promote rigor and ethics must itself be exemplary in how it operates. IOITE's governance follows three principles: institutional sobriety — creating a body only when it becomes necessary; transparency — making processes and outputs public; and prevention of conflicts of interest, particularly with respect to commercial vendors from whom we intend to remain independent.

The initial legal form is that of an association, with bylaws, a general assembly and a board of directors. This choice — a French "loi 1901" association or an association under Articles 60 et seq. of the Swiss Civil Code — offers a lightweight legal basis, with no minimum capital, suited to an emerging movement. It does not prejudice later, more ambitious forms the organization may take if its recognition broadens; but it constitutes a realistic and immediately operational starting point. The precise choice of jurisdiction, which carries tax and liability consequences, must be settled with the support of a legal professional.

The general assembly brings together the members and holds sovereignty over direction, accounts, and the election of the board of directors. The board administers the association and elects from among its members an executive committee — chair, vice-chair, secretary-general, treasurer. Membership composition reflects the project's gradual maturity: founding members, present before the first broader public recognition; accredited members, who will meet competence and ethics criteria once the certification framework is operational; associate members, individuals or organizations supporting the mission without yet meeting accreditation criteria; and honorary members, invited in recognition of their contribution to the field. This gradation matters: it allows us to welcome support today, without pretending to accredit members against criteria that do not yet exist. One does not grant accreditation before defining what it attests to.

Substantive work is entrusted to thematic councils — specialized working groups on ethics, cybersecurity, artificial intelligence, cloud, green IT, education, architecture — tasked with producing public recommendations subject to board approval. These councils are the organization's intellectual engine; it is through the quality of their work, following the example of the major technical organizations mentioned above, that our legitimacy will be built. An observatory function is attached to them, tasked with regularly publishing data on the state of the profession.

Two bodies are intended to come into existence only at an advanced stage of maturity, and we want to be clear about this so as not to suggest powers we do not exercise. The member accreditation mechanism presupposes that the Competency Framework is

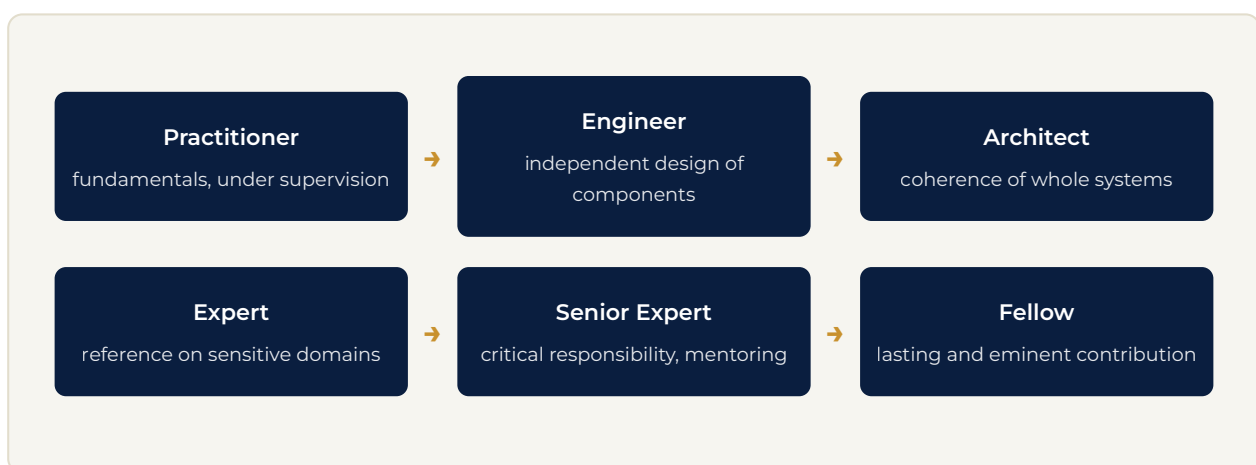
finalized; it will only be activated at that point. The disciplinary chamber, tasked with investigating breaches of the code of ethics among accredited members, will only be able to function once the certification and accreditation framework is itself mature and beyond dispute. Instituting discipline without a solid foundation would be illegitimate; we rule it out. Finally, in all its communications, the organization must clarify its associative status and avoid any confusion with a legally constituted professional order — a rule our bylaws state explicitly.

## 10. Certification Framework and Levels

Certification is the most delicate workstream, because it is where the risk of reproducing the very flaws we criticize is greatest. The market is already saturated with certifications; many attest to mastery of a particular commercial product rather than a transferable, durable professional competence. Our ambition is precisely to offer an independent certification, in the sense of our glossary: issued by the organization or its academic partners, and not tied to a commercial software vendor.

Before detailing this scale, a common confusion needs clearing up — one that was in fact flagged to us by a reviewer from outside the field: teaching coding basics in middle school, now widespread in many education systems, does not make every pupil an IT professional, any more than teaching science at school makes every pupil an engineer. We therefore distinguish four quite different tiers. **Digital literacy** is the general ability to understand and use digital technologies — an education goal for everyone, not a professional status. The **IT practitioner** carries out technical activities within a defined scope, generally under supervision. The **professional** regularly performs IT-related work with identified responsibilities. The **IT expert**, finally, is the professional capable of handling complex situations, exercising independent judgment, demonstrating its validity, and bearing the consequences — the definition set out in this document's executive summary.

The foundation of this framework is a scale of expertise levels, which our glossary sets as follows: Practitioner, Engineer, Architect, Expert, Senior Expert, and Fellow. This progression is not a hierarchy of prestige but a gradation of responsibility and autonomy, which can be read as the professional life cycle of the IT expert: at each rung, knowledge, demonstrated skills and verifiable experience deepen.





The Practitioner masters the fundamentals and works under supervision; the Engineer independently designs and implements components; the Architect assumes the coherence of whole systems and structural trade-offs; the Expert and Senior Expert carry reference-level responsibility over sensitive, including critical, domains, as well as for mentoring; the Fellow is a recognition of durable and eminent contribution to the discipline. Each level will need to be precisely defined in the Competency Framework, associating with each rung specific knowledge, demonstrated skills and verifiable experience.

Three principles must govern this framework so that it does not become just another certification. The first is independence: the design of frameworks and evaluation must be kept apart from commercial interests, which presupposes academic backing and strict rules to prevent conflicts of interest. The second is validity: a certification is only worth something if it actually measures what it claims to measure. This means favoring the assessment of demonstrated competencies — case studies, reviews of past work, situational exercises — over the mere testing of memorized knowledge, and subjecting the framework itself to regular critical evaluation. The third is transparency: the frameworks, criteria and processes must be public, so that the value of the certification can be assessed by everyone — employers, peers, the public.

We want to be explicit about one limitation, out of honesty. As long as IOITE has no legal recognition — and we have said it has none — any certification it issues has no legal value; it does not condition access to any trade and confers no protected title. Its value, initially, will be purely that of a quality signal, following the model of standards produced by organizations with no binding power but universal respect. This is a deliberate choice: we prefer a certification with no legal value but genuine credibility, over a grand-sounding title with no substance. Credibility will be built through the rigor of the framework and through the recognition it earns, voluntarily, from universities and employers. It is only at the end of this path that a more formal recognition could, eventually, be discussed.

## 11. Mandatory Continuing Education

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Of all our proposals, the requirement for continuing education is perhaps the least contestable, since it follows directly from the nature of the trade. Few professions see their knowledge renewed as fast as IT. Languages, architectures, threats, regulatory frameworks appear and transform within a few years. A professional who stopped training would see their competence erode quickly — not for lack of talent, but through the obsolescence of knowledge. In this, IT resembles medicine, where continuing education has long been a recognized obligation, precisely because people's safety depends on up-to-date knowledge.

We therefore propose that continuing education, in the sense of our glossary — an obligation to regularly update one's competencies, formalized in annual hours per domain — become a pillar of accredited-member status, once that status is operational. We envisage a breakdown by major domain: artificial intelligence, cybersecurity, cloud and infrastructure, regulation, ethics, and digital sustainability. The goal is not to count hours for their own sake, but to ensure that a professional claiming a given level of expertise actually maintains the corresponding knowledge. Continuing education is the natural counterpart to certification: the latter attests to a competence at a given moment, the former guarantees it does not go stale.

This requirement must be designed realistically, so as not to become a bureaucratic burden disconnected from practice. Three safeguards seem essential to us. First, the diversity of recognized formats: continuing education is not limited to formal courses; it includes participation in standardization work, contribution to open projects, publishing, mentoring, structured monitoring of the field. A profession that learns a great deal through practice and peer sharing must value these forms. Second, content independence: as with certification, continuing education must not become a disguised channel for commercial promotion; content tied to a specific product can only make up a limited share. Finally, proportionality: the requirement should be higher for those working on critical systems, and lighter elsewhere, consistent with the targeted principle governing our whole approach.

As long as the accreditation framework is not operational, this requirement will remain a recommendation and a voluntary commitment, not an enforceable obligation — we could not impose what we do not yet have the mandate to oversee. But we intend, from now on, to lay down its principles, document reference pathways, and lead by example within the organization itself.

## 12. Intended Partnerships

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An organization like ours cannot succeed alone; its legitimacy will be built through relationships with three groups of actors, whose voluntary buy-in will be the best indicator of our credibility. We approach these partnerships in a spirit of cooperation, not capture: we seek to complement what already exists, not to compete with it.

The first partners are **universities and schools**. They are the ones training future professionals, and they hold the academic independence needed to design competency frameworks and evaluation systems shielded from commercial interests. A partnership with the academic world is the very condition of our certification framework's credibility. We wish to work on this concretely: co-building frameworks, giving evaluations a scientific grounding, articulating with existing curricula and degrees — not to replace them, but to offer a shared language linking initial training and professional practice. Recognized institutions in computer science and engineering, in French-speaking Europe and beyond, form our natural first circle of interlocutors.

The second partners are **companies**, particularly those that design or operate critical systems. They have a direct interest in our approach: a common framework and shared ethics reduce information asymmetry in the labor market, make hiring safer, and give recognition to professionals who invest in their competence. Chief information security officers and technical leadership, for their part, provide the operational grounding without which our frameworks would remain theoretical. We propose to involve them through voluntary commitments — a charter for partner companies — less binding than a code of ethics, but carrying mutual recognition.

The third partners, over the longer term, are **institutions and public authorities**. We address them last, and deliberately so. It would be presumptuous, and contrary to the honesty we claim, to seek institutional recognition before having proven our seriousness. Our strategy is the reverse: first build a solid corpus and credible academic and professional partnerships, then, only then, contribute to existing public consultations on digital policy and put forward our expertise. Institutional recognition, if it comes, will be the consequence of our credibility, not its precondition. We approach public authorities as a possible endpoint, never as an endorsement we would claim for ourselves.

Across all these partnerships, one rule holds: recalling our actual status. We are an associative movement, not a legal order, and we only engage our partners on the basis of a free, informed choice.

## 13. Roadmap

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Our roadmap is deliberately designed around sequential milestones rather than calendar deadlines. A founding movement does not progress at the pace of an imposed calendar, but at the pace of the actual maturity of its output and the buy-in it generates. Announcing dates that nothing guarantees would harm our credibility; we prefer to describe a logical sequence of steps, each conditioning the next.

The first milestone is legal and identity lock-in: constituting the association, securing the name and domain names, filing trademarks in relevant jurisdictions, reserving a coherent online presence. This is the minimum legal building block, the one that gives the movement formal existence and protects its name.

The second milestone is establishing a minimum viable documentary base: a locked-down glossary, a clear vision, mission and manifesto, the executive summary of this White Paper ready for immediate publication, and the association's bylaws. This base already largely exists; the present complete White Paper is its natural extension. Next comes the public presence milestone: a first-version website, distribution of the executive summary, and factual, rigorous communication free of controversy — since our authority will be built as much through the sobriety of our discourse as through the solidity of our substance.

The fourth milestone is the complete corpus: beyond the White Paper, the code of ethics, the Competency and Level Framework, and the charters intended for members, companies and universities. This is the intellectual foundation on which all future recognition will rest. The fifth milestone consists of assembling a first circle of credible contributors — academics, security officers, technical leaders, digital-law specialists — approached first for opinions and occasional reviews, never for immediate membership. Setting up an informal advisory council will precede any permanent structure.

The final milestone is broader recognition: formal partnerships with universities and schools, participation in existing public consultations on digital policy, and the publication of a regular report on the state of the profession through the observatory function. It is only at this stage that the question of a more formal institutional recognition can legitimately be raised, in countries where the legal framework allows it. We stress the order of these milestones: each presupposes the previous one. Recognition cannot be built before the corpus, nor the corpus before a shared vocabulary. This discipline in progression is, in itself, a guarantee of seriousness.

## 14. Conclusion and Call to Action

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In this White Paper, we have tried to hold together two requirements that might seem contradictory: ambition and modesty. Ambition, because the diagnosis is serious and deserves a response to match — software has become critical infrastructure for our societies, and the profession that produces it lacks the structuring that this responsibility would call for. Modesty, because we know what we are and what we are not. We are not a professional order in the legal sense; we do not have the power to be one, and no private organization can self-attribute this. We are a founding movement that proposes, documents and builds — patiently, openly, with no claim to an authority we do not hold.

Our proposal is targeted: not to regulate all digital trades uniformly, but to ensure that those who bear technical responsibility for critical systems can rely on a shared competency framework, a common code of ethics, an independent certification and continuing education. It is progressive: it advances through milestones, each conditioning the next, without skipping steps or promising what it cannot deliver. It is open: it seeks neither monopoly nor artificial barriers, but better quality, transparency and trust — for the benefit of professionals as much as of society.

To those who might doubt the usefulness of this approach, we recall the textbook cases discussed in this document: they are not isolated accidents but the symptoms of a collective blind spot — that of a profession whose effects on the world have changed scale without its structuring keeping pace. To those who fear the trade would become closed, we answer that IT's own history — its openness, its creativity, its inclusion of talent from all backgrounds — is an asset we intend to preserve, and that our requirement applies only to a precise scope, where the social cost of having no guarantees whatsoever has become plain.

This document is a version 3, and it will remain, by methodological choice, a work in progress for a long time: we want it to be criticized, corrected and enriched before it is treated as settled. We therefore call for contributions from academics, information-system security officers, digital-law specialists, engineers and field practitioners. Whether you disagree with our diagnosis, are reserved about our proposals, or are ready to contribute, your voice is valuable to us. Competence, ethics, responsibility: these three words are not a slogan, but a program of work. This project is not an endpoint — it is a beginning, and it belongs to you as much as to us.

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# The 10 Commitments of the IT Expert

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A condensed version of the code of ethics, intended for quick reading. Each commitment is developed in the full code of ethics; in case of doubt, the latter prevails.

1. **I place the safety of people and the public interest above any other interest**, including commercial or hierarchical, when they conflict.
  2. **I only practice within the limits of my actual competence**, and I honestly disclose what I do not master rather than concealing it.
  3. **I protect the data entrusted to me**, its confidentiality, its integrity, and the privacy of the people concerned.
  4. **I design security into a system from the outset**, not as a fix bolted on afterward.
  5. **I document my important technical decisions**, their assumptions and their limitations, so that they can be understood, verified and revisited.
  6. **I disclose my conflicts of interest** and refuse any fraudulent, misleading or concealed practice.
  7. **I answer for my technical choices** and never hide behind a tool, a machine or an automated system to escape my responsibility.
  8. **I design lean systems**, mindful of their environmental footprint and the fair use of resources.
  9. **I maintain my competence through regular continuing education**, aware that IT knowledge ages quickly.
  10. **I pass on my knowledge and respect my peers**, contributing to raise the collective standard of the profession rather than protecting an individual advantage.
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# Founding Member Charter

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A voluntary commitment, expressed in the first person, undertaken by anyone joining IOITE as a founding member. This charter has no enforceable legal value; it is a moral commitment and a shared statement of intent.

By joining the International Order of IT Experts as a founding member, I commit to:

- **Serve the mission** of the movement: contribute to a progressive, targeted and honest recognition of the IT profession, for the benefit of society as much as of professionals.
- **Respect the truth about our status:** never present IOITE as a legally constituted professional order, and recall, whenever necessary, that it is a founding movement backed by an association.
- **Let quality prevail over posturing:** build our legitimacy through the rigor of our work and the transparency of our processes, not through titles or claims to authority we do not have.
- **Defend the openness of the profession:** promote requirements where they are necessary — on critical systems — while never seeking to erect artificial barriers or protect entrenched positions.
- **Embody the ethical commitments** we propose to the profession, leading by example in my own practice, including in continuing education and digital sustainability.
- **Prevent conflicts of interest,** particularly with respect to commercial vendors, and preserve the organization's independence.
- **Contribute in good faith** to the shared corpus — through review, constructive criticism, production or dissemination — in a spirit of cooperation and respect for peers.
- **Welcome disagreement:** treat criticism and dissent as raw material for the project, not as a threat.

I subscribe to these commitments freely, aware that they bind me on my honor and that they reflect the spirit in which this movement was founded.

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## Annex A — State of the Profession Around the World

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This annex offers a deliberately qualitative comparative overview of how the IT profession is structured in a few countries. The information below reflects a general and evolving situation, to the best of our knowledge and as of the date of writing; it does not constitute legal advice and should be verified and updated with the support of local specialists. The common thread across all these countries deserves to be highlighted upfront: nowhere is IT, to date, a fully regulated profession — with a protected title and reserved acts of its own — as medicine or law are.

**France.** IT is not a regulated profession there: no title or diploma is legally required to design or operate most systems. The title of "chartered engineer" (ingénieur diplômé) is overseen by the Commission des titres d'ingénieur (CTI), but it protects an academic qualification, not a professional act reserved to IT. Professional and trade organizations for the digital sector exist (trade federations, learned societies), without ordinal character. In specific areas, the law imposes functions or obligations — for example the data protection officer under the GDPR — but these are targeted regulatory roles, not a status for the profession as a whole.

**Switzerland.** The IT profession is likewise not regulated in the ordinal sense. Training relies on a robust dual system and on qualifications awarded by polytechnic schools and universities of applied sciences; professional associations and trade bodies structure continuing education and trade certifications. As in France, there is no legally protected title of IT expert, nor an order with disciplinary power over the profession as a whole.

**Canada (and Quebec).** Canada offers the most developed example of regulated engineering. In Quebec, the Ordre des ingénieurs du Québec protects the title of engineer, provided for by law, with reserved activities in certain fields and a disciplinary mechanism; the tradition of the Iron Ring symbolically anchors responsibility there. Software engineering has been partially integrated into this framework, depending on the province. There are also professional designations specific to IT carried by associations, but they generally do not confer the same legal status as the engineering title. No order specifically covers non-engineer IT professionals as a whole.

**United States.** Access to the profession is largely open. The Professional Engineer license is issued at the state level and mainly concerns physical engineering; a licensing exam in software engineering existed before being discontinued for lack of sufficient candidates — an illustration of the resistance to transposing the classical engineering model onto



software. Structuring instead comes mainly through leading learned societies (IEEE, ACM), whose authority is intellectual and voluntary, and through a bustling market of private certifications, often tied to vendors.

**United Kingdom.** The profession is not regulated by law, but it has a relatively developed voluntary-recognition system. A leading professional institution, holding a royal charter, awards professional statuses (for example, chartered IT professional recognition), and a path to chartered engineer status exists via the relevant national engineering body. These recognitions reward competence and ethical commitment, but they rest on voluntary participation and do not legally condition practice.

**Germany.** The title "engineer" (Ingenieur) enjoys a form of protection, with variations by Land, mainly inherited from traditional engineering. The IT profession itself is not regulated in the ordinal sense; it rests on a recognized dual training system and on an influential national learned society for computer science, which contributes to frameworks and ethics in the discipline without exercising general disciplinary power.

**What this overview teaches us.** Despite very different institutional traditions, none of these countries has given IT a complete professional framework — protected title, enforceable code of ethics, discipline and mandatory continuing education — comparable to that of the health or legal professions. Where structuring exists, it follows either the path of regulated engineering (Canada) or that of voluntary recognition by learned institutions (United Kingdom, United States, Germany). This finding reinforces our approach: start by building, at an international scale and through voluntary buy-in, authority through quality — before considering, country by country and only where the legal framework allows it, a more formal recognition.

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## Annex B — Bibliography and References

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This annex lists the standards, organizations and regulatory frameworks cited in the body of the White Paper. Each entry gives its official name and a brief description. The standard and regulatory identifiers mentioned below are stable public references; readers are invited to check with the issuing bodies for the version in force at the time of reading, as standards and regulations are subject to periodic revision.

### *Organizations and Learned Societies*

- **IEEE** — Institute of Electrical and Electronics Engineers: leading international learned society in electrical, electronic and computer engineering, publisher of standards and scientific publications.
- **ACM** — Association for Computing Machinery: international learned society dedicated to computer science, its research and its teaching.
- **IETF** — Internet Engineering Task Force: open organization that develops and publishes the Internet's technical standards in the form of RFC (Request for Comments) documents.
- **W3C** — World Wide Web Consortium: international standards body for Web technologies (HTML, CSS, related protocols and formats).
- **OWASP** — Open Worldwide Application Security Project: open, non-profit community producing recommendations, tools and frameworks for application security.
- **Ordre des ingénieurs du Québec (OIQ)** — professional order governed by Quebec's Code des professions, which oversees the legally protected title of engineer, its reserved activities and its discipline.

### *Reference Ethics Texts*

- **IEEE-CS/ACM Software Engineering Code of Ethics and Professional Practice** — code of ethics and professional practice for software engineering, jointly developed by IEEE Computer Society and ACM; an international reference, though without binding legal value.

### *Technical, Security and Safety Standards*

- **ISO/IEC 27001** — international standard on information security management systems (ISMS), published by ISO and IEC.

- **DO-178C** — Software Considerations in Airborne Systems and Equipment Certification framework, governing embedded avionics software and imposing rigorous traceability between requirements, code and tests.
- **IEC 62304** — standard governing software lifecycle processes for medical devices.
- **ISO 26262** — standard addressing the functional safety of electrical and electronic systems in road vehicles.
- **IEC 61508** — generic functional-safety standard for safety-related electrical, electronic and programmable electronic systems, the foundation of a family of derived sector-specific standards.
- **CVE / CVSS** — Common Vulnerabilities and Exposures (a public, standardized catalogue of security vulnerabilities) and Common Vulnerability Scoring System (a public system for scoring their severity).

### *European Union Regulatory Frameworks*

- **GDPR** — General Data Protection Regulation, Regulation (EU) 2016/679, on the protection of natural persons with regard to the processing of personal data.
- **NIS2 Directive** — Directive (EU) 2022/2555, raising the common level of cybersecurity and resilience of network and information systems for essential and important entities in the Union.
- **DORA** — Digital Operational Resilience Act, Regulation (EU) 2022/2554, on the digital operational resilience of the financial sector.
- **Cyber Resilience Act (CRA)** — European Union regulation on cybersecurity requirements for products with digital elements, adopted in 2024.
- **Artificial Intelligence Act (AI Act)** — Regulation (EU) 2024/1689, establishing harmonized rules on artificial intelligence following a graduated, risk-based approach.

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This White Paper constitutes a version 3.1 consolidation, prepared following an external review of version 3. It now includes, in addition to the author's personal preface, the About the Author section, the About IOITE section (vision, mission, values), the standalone executive summary, explanatory diagrams, the condensed version of the code of ethics, the founding member charter, the international overview and the reference bibliography: a clarification of the two-tier structure between the legal entity and the movement as well as of the international ambition (About IOITE section), an explicit operational definition of the IT expert (executive summary and Section 10), an explicit hierarchy of purposes placing the protection of the public ahead of professional recognition (Section 8), a distinction

between digital literacy and professional expertise (Section 10), and nuances of causality regarding cybercrime (executive summary) and the causes of major outages (Section 3), to avoid any promise or shortcut the facts do not support. The terms used follow the definitions in IOITE's founding Glossary, a reference document readers are invited to consult.

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# Software has become a matter of public interest.

Every day, millions of people entrust their savings, their health, their communications, their mobility and, at times, their safety to invisible software systems. Yet even as these technologies are held to ever stricter standards, the practice of the IT profession itself remains largely without a shared code of professional ethics.

This White Paper reflects on the place of IT in our modern societies and lays the foundations of an international movement to promote competence, ethics and professional responsibility in the design of digital systems.

More than a reform, this is a collective awakening: as software becomes essential infrastructure, those who design it must fully assume the responsibility that comes with it.

**COMPETENCE. ETHICS. RESPONSIBILITY.**



## YANN DEUNGOUÉ

### SOFTWARE ARCHITECT & TECHNICAL LEAD

With over 15 years of experience in finance, critical systems and enterprise architecture, he works to raise the standards of the IT profession.

Founder of the international IOITE movement, he leads this initiative with the ambition of structuring the profession to match its real responsibility in critical systems.



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